



*Better study,
better decisions.*

Kogilu Site Report

Site 3 · Sy. 27 & 28 · 10.84 acres

SITE ANALYSIS

Prepared for Qnit case-study portfolio
Presented by GeoAce Studio Pvt Ltd

Kogilu, Yelahanka, Bengaluru North · 13.098754° N, 77.609015° E
10 Oct 2026 / Rev A / QNIT-BLR-S03





The site at a glance

Site 3 — Kogilu / Kogilu, Yelahanka, Bengaluru



LATITUDE

13.098754° N

LONGITUDE

77.609015° E

STUDY EXTENT

346 m

North up · scale bar on image
Imagery Wayback 24 Sep 2026

SOURCE / Esri World Imagery (Wayback 2026-09-24). Boundary from Case_Studies.kml, Site 3; vertices V1–V4, centroid in WGS 84.

SITE AREA

43,886 m²

10.84 acres · 4.39 ha · 434 guntas

PERIMETER

856 m

KML boundary, Oct 2026

BOUNDARY

4 edges

Quadrilateral, about 259.0 × 201.0 m

Design brief

A 4.39 ha residential parcel in Kogilu, north Bengaluru, already subdivided as a plotted layout. What can it carry under RMP 2015 Residential (Main) controls, and which edge, buffer and height clearance sets the limit?

Primary opportunity

99.7% of the parcel reads Residential on the draft RMP 2031 layer; Residential (Main) on approved RMP 2015. A school is a 5-min walk and Yelahanka metro (Blue Line 2B) is due 2027–28: group housing at FAR 2.25–2.50 is the natural programme.

Key constraint

A mapped stream runs 5 m off the west edge (D–A) in the valley "sensitive area". A 25–50 m drain buffer takes 1,574–5,750 m² (3.6–13.1%). Buildings of 15 m or more need Yelahanka AFS clearance.

REPORT ROUTE

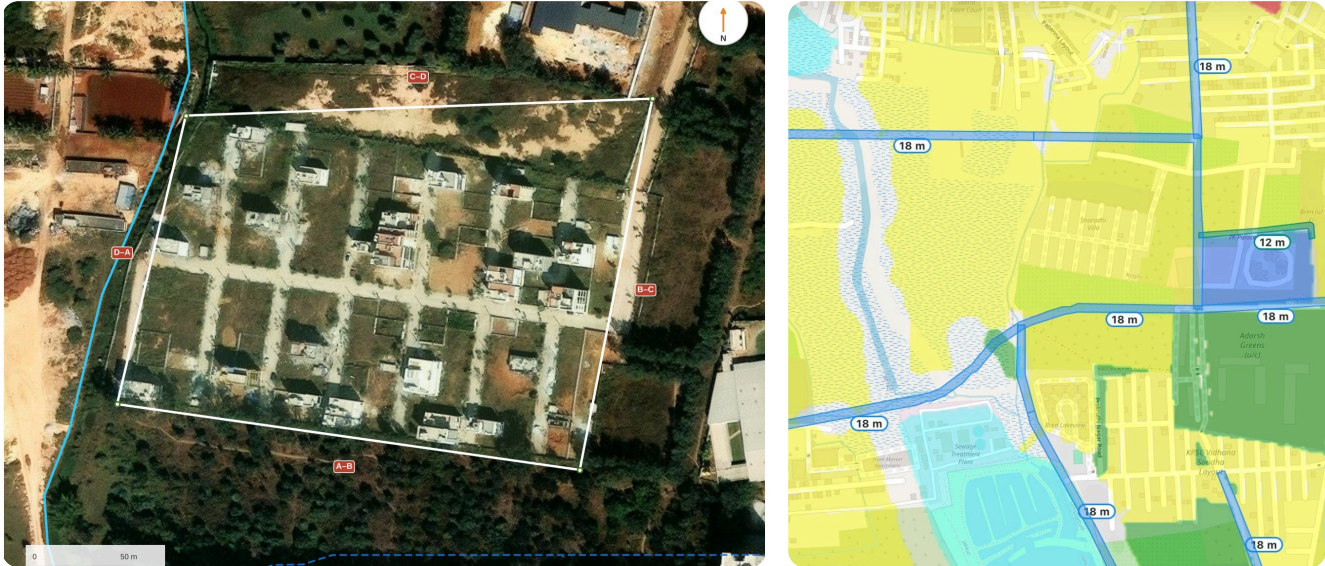
03–04 Site & ground 05–08 Environment
09–11 Context & services 12 Priorities & sources

Stakeholders: Qnit case-study portfolio / GeoAce Studio design team / Reviewer: pending



Boundary, edges & development

Site 3 — Kogilu / Kogilu, Yelahanka, Bengaluru



SOURCE / Left: Esri imagery 2026; stream from Overture Maps (2026-09); dashed = draft RMP 2031 plan roads (Qnit planning layer). Right: Qnit 2031 plan layer, BDA RMP 2031 (draft), supplied screenshot.

Site identity

Survey / parcel **Sy. 27/*1 · 28/*IXX**
Village / authority **Kogilu (938519) · BDA**
CRS / boundary **WGS 84 KML · UTM 43N**
Records / status **RCCMS: no cases · no mutations**

Development controls

Permitted use / source **Residential (Main) · RMP 2015**
FAR / coverage **2.25–2.50 / 55% (18 m road)**
Setbacks F / R / sides **≥ 5 m all round; 5–16 m by height**
Height limit / basis **< 15 m without AFS NOC**

Edge conditions

Edge	Adjoining condition	Clearance / width	Evidence
A–B · S	Tree belt (plantation strip), open land; Jakkur Lake beyond	Jakkur Lake 419 m S; plan 18 m road 45 m SE	Esri 2026 · Overture · RMP 2031 layer
B–C · E	Local residential road; open plots and low sheds	Road width not mapped; survey needed	Overture 2026-09 · Esri 2026
C–D · N	Vacant plot with earth track; 1–2 storey houses beyond	No frontage road	Esri 2026 · Overture buildings
D–A · W	Earth track, mapped stream in the valley strip; brick and sand yards beyond	Stream 5 m; primary drain 284 m SW	Overture water · BBMP SWD 2022

DESIGN / DEVELOPMENT IMPLICATION

The west edge controls. RMP 2015 sets 15 / 25 / 50 m drain buffers by drain class: here that is 394 / 1,574 / 5,750 m² of the site. Confirm the drain class with GBA (SWD) and the revenue sketch, and the widths of the E and W roads, before fixing the layout.



Terrain & the ground beneath

Site 3 — Kogilu / Kogilu, Yelahanka, Bengaluru



SOURCE / Copernicus GLO-30 DSM (30 m, resampled to 10 m), 1 m contours, over Esri imagery. A–B follows the fitted fall line.

ELEVATION RANGE

891 / 900 m

EGM2008 · DSM includes canopy

TYPICAL SLOPE

2.2 %

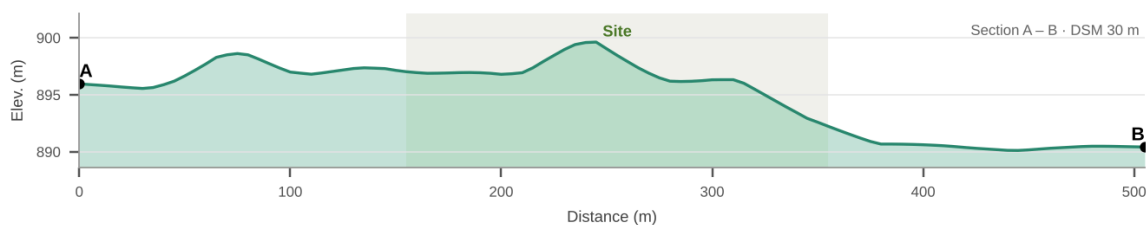
Gentle · falls to the NNW

CONTOUR INTERVAL

1 m

GLO-30 DSM / 30 m

Transect A–B



SOURCE / Section A–B sampled every 5 m from the DSM; A is south-east, B north-west.

Section length 506 m / Relief 9.5 m / Vertical exaggeration ≈ 4×

DESIGN / DEVELOPMENT IMPLICATION

About 9 m of fall across the parcel, 2.2% to the NNW; the low point (891 m) is the north-west corner, where runoff leaves for the western valley. Step block platforms by 0.5–1 m; no tall retaining walls are indicated. Confirm with a ground survey, as the DSM includes trees.



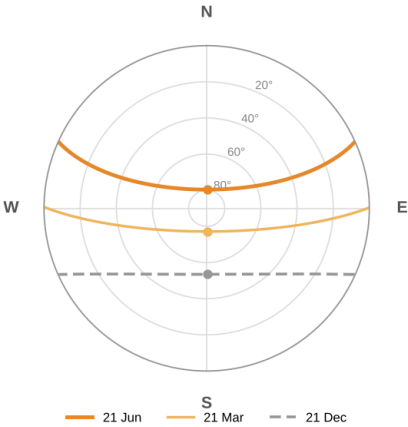
Light, shade & orientation

Site 3 — Kogilu / Kogilu, Yelahanka, Bengaluru

01 / SUN PATH



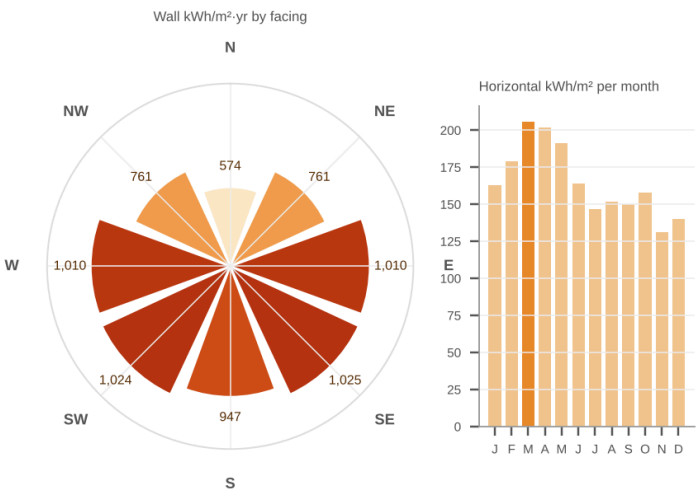
SEASONAL COMPARISON



21 Mar · 21 Jun · 21 Dec; IST (UTC+5:30)
True north; noon altitude 54° (Dec) to 80° (Jun, sun to the north)
Day length 11.4–12.9 h; no context model

SOURCE / Solar positions computed (NREL SPA, pvlib) for the site centroid; stereographic dome, Esri imagery. Hour lines at the 21st of each month.

02 / SOLAR RADIATION



ANALYSIS PERIOD

2015–2024 monthly means

INCIDENT SOLAR ENERGY

1,983 kWh/m²

Horizontal plane, per year (5.43 kWh/m²·day)
Highest wall exposure: SE / SW ≈ 1,025; north 574
Best month Mar; cloud cuts 17% vs clear sky

SOURCE / NASA POWER CERES SYN1deg monthly GHI 2015–2024; wall split modelled from clear-sky geometry scaled to it.

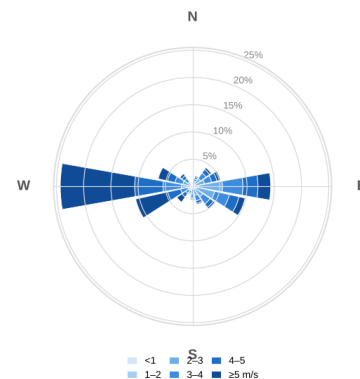
DESIGN / DEVELOPMENT IMPLICATION

The long axis runs E–W (98°), so blocks can face north and south: the noon sun stays above 54°, and short overhangs shade both faces. East and west walls take ≈ 1,010 kWh/m²·yr: limit glazing there. Roofs suit PV.



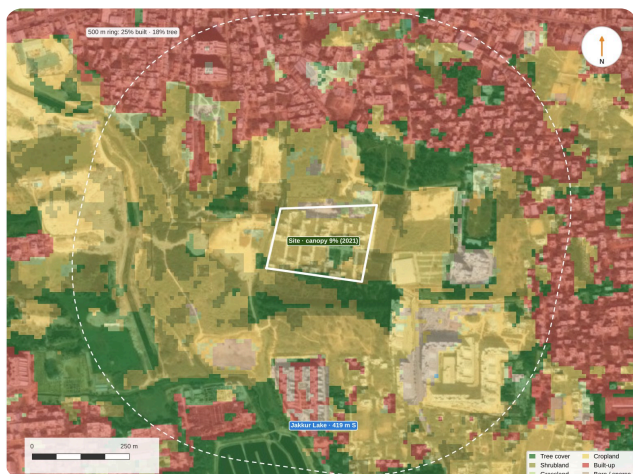
Air movement & outdoor comfort

Site 3 — Kogilu / Kogilu, Yelahanka, Bengaluru



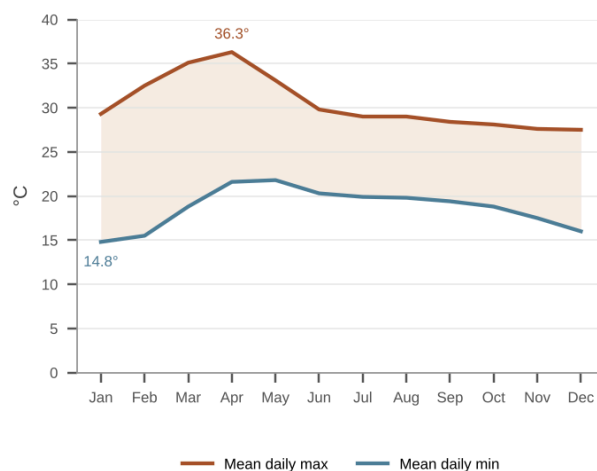
SOURCE / NOAA ISD hourly observations, Kempegowda Intl Airport (VOBL), 15.2 km NE; agrees with NASA POWER MERRA-2 (from W, 3.9 m/s).

Local comfort



SOURCE / ESA WorldCover 2021 (10 m) on Esri imagery; dashed = 500 m ring.

Seasonal temperature



SOURCE / NASA POWER MERRA-2 daily, 2015–2024. VOBL station mean 24.2 °C vs model 24.0 °C.

DESIGN / DEVELOPMENT IMPLICATION

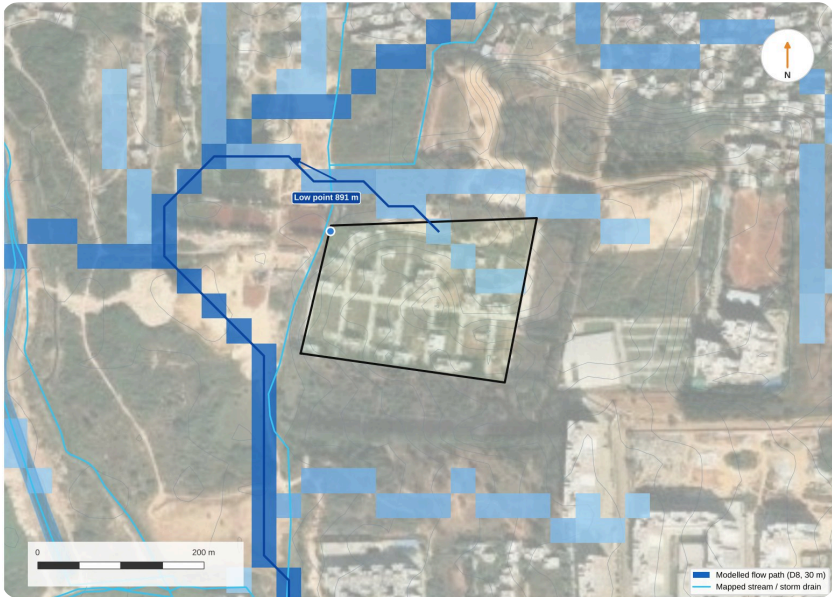
Run internal streets and gaps E–W so the June–September westerly (5.1 m/s) flushes the layout, entering over the open west valley. April is the hot month (mean max 36.3 °C): keep the southern tree belt and shade play areas and parking.



Runoff, storage & groundwater

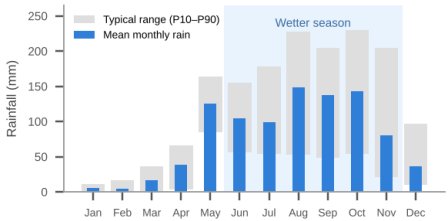
Site 3 — Kogilu / Kogilu, Yelahanka, Bengaluru

SURFACE RESPONSE



SOURCE / D8 flow on GLO-30 DSM; Overture water; BBMP primary storm drains (SWD 2022). Rain: NASA POWER MERRA-2 grid 13.0 N 77.5 E.

MONTHLY RAINFALL

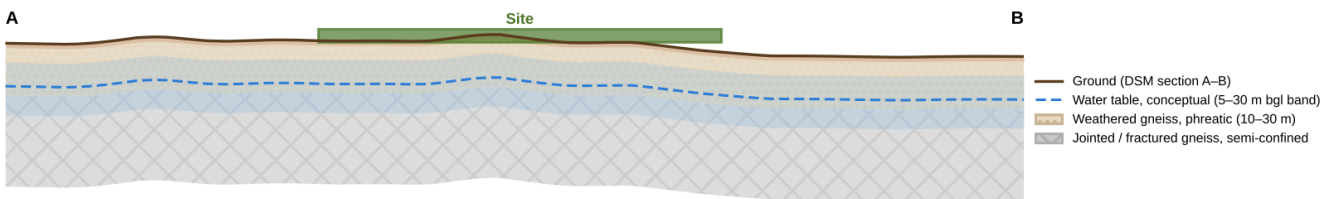


EVENT BASIS

95 mm / 24 h

10-year return period (Gumbel on 1981–2025 annual maxima)
Mean annual 937 mm; 65% falls May–Nov

Groundwater & recharge



SOURCE / Conceptual section on A–B. Strata and water depths from the CGWB Bengaluru Urban profile, not site boreholes.

Surface water

Low point NW corner (891 m) → west stream → primary drain
284 m SW → Jakkur Lake.
10-yr day on 4.39 ha: $\approx 1,500 \text{ m}^3$ runoff now (C 0.35), $\approx 3,100 \text{ m}^3$ built (C 0.75).
Basis: catchment = site; land cover WorldCover 2021.

Ground conditions

District water level 5–30 m bgl pre-monsoon (CGWB).
Bengaluru North extraction over 200% of recharge (2024): over-exploited.
Weathered gneiss 10–30 m over fractured rock; percolation to be tested.

DESIGN / DEVELOPMENT IMPLICATION

Keep the NW corner and west edge as the drainage and detention zone, sized for $\approx 1,700 \text{ m}^3$ of extra runoff once built; do not culvert the stream. Plan water on Cauvery Stage V or treated reuse, not new borewells.



Living systems & views

Site 3 — Kogilu / Kogilu, Yelahanka, Bengaluru



SOURCE / Esri imagery; Overture water bodies; plan-view sight lines from the site centre to the three nearest named lakes.

Landscape structure



Landform



Vegetation



Built form

Canopy cover **9% site · 18% 500 m ring**
 Permeable ground **100% (2021) · 75% ring**
 Water / habitat edge **stream 5 m · Jakkur Lake 419 m**
 Tree survey / season **none · WorldCover 2021**

View & habitat priorities

Retain

The southern tree belt along A–B and the west valley corridor linking to Jakkur Lake.

Investigate

Lake views are screened at ground level; upper floors gain them, but heights of 15 m or more need AFS clearance.

Observer: site centre, plan view (no eye-height model)
 Vegetation 2021 predates the layout roads; built cover now higher

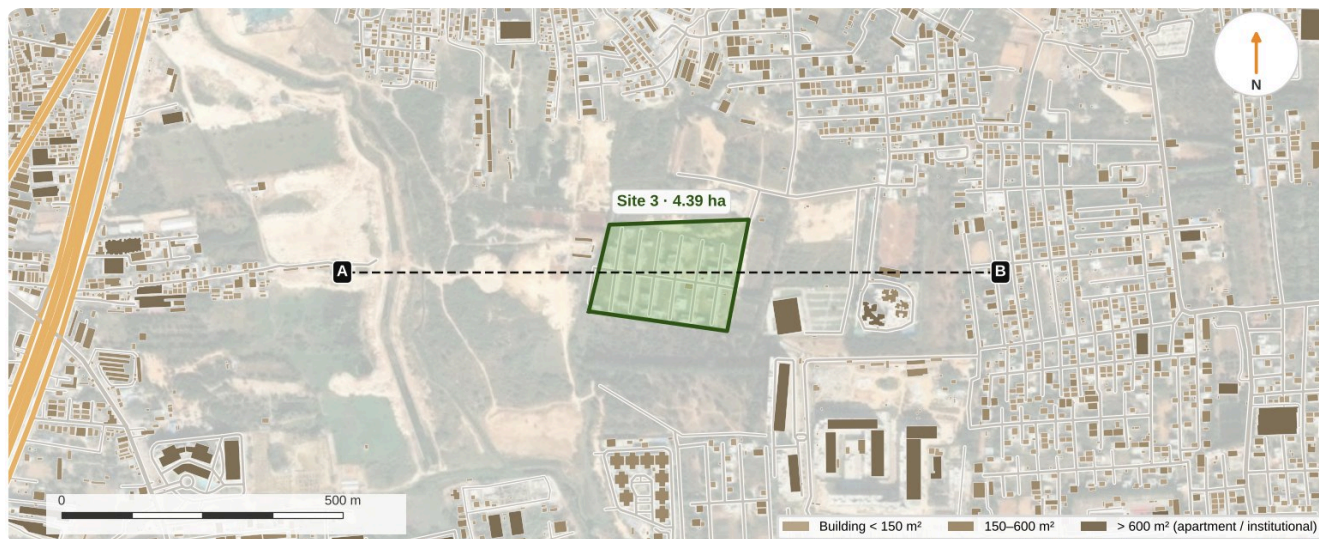
DESIGN / DEVELOPMENT IMPLICATION

Put the statutory 10% park (4,389 m², RMP 2015 large-plot rule) along the west valley, so it doubles as the drain buffer and the green link to Jakkur Lake.



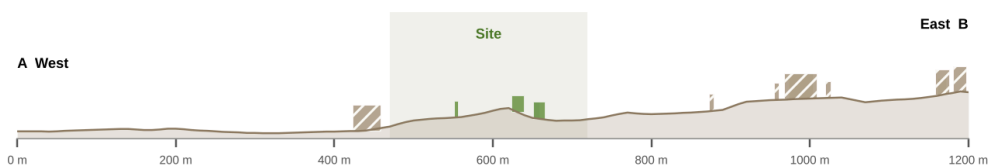
Neighbourhood fit & movement

Site 3 — Kogilu / Kogilu, Yelahanka, Bengaluru



SOURCE / Overture Maps buildings and roads (release 2026-09-23) over Esri imagery; A-B is a 1.2 km W-E section through the site.

STREET & SKYLINE PROFILE A-B



13 buildings cut; none carry mapped heights, so hatched blocks use storeys assumed from footprint (2-4 floors × 3.2 m).

DAILY ACCESS



Walking 5 / 10 / 15 min at 4.8 km/h on the Overture road network from the layout roads.

ACCESS BASIS

Layout roads / walk / weekday

Transit **Spathagiri Layout stop** · 545 m
 Rail **Yelahanka Jn** · 1.9 km; metro due 2027-28
 Health **Sri Maruthi Hospital** · 14 min
 Education **Anand Vidhya** · 5 min
 Daily needs **Spar Hypermarket** · 23 min

No mapped road on the N and S edges; access relies on the E and W lanes. NH-44 (Bellary Rd) is 0.9 km W.
 Context is 1-3 storey houses (median footprint 75 m²) with 9 large blocks: mid-rise up to 15 m fits.

Overture 2026-09-23.0 · checked 10 Oct 2026



The context beyond the boundary

Site 3 — Kogilu / Kogilu, Yelahanka, Bengaluru

LAND-USE CHANGE

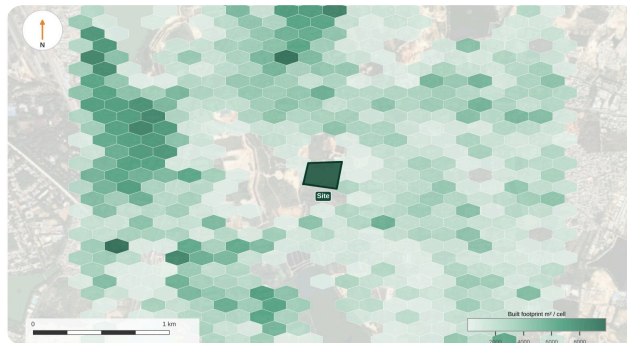


SOURCE / Esri Wayback releases 27 Feb 2017 (left) and 24 Sep 2026 (right).

Recorded change

2017: orchard with tracks → 2026: plotted layout, road grid and about 20 houses.
Imagery 2017 / 2026; WorldCover 2021: 64% cropland.
No heritage mapped on site; waste site planned 2.4 km NE.

PEOPLE & ACTIVITY



SOURCE / Built-footprint density from Overture buildings, 2026; activity proxy.

Community needs

Census 2011 figure not published for Kogilu; 921 buildings within 500 m.
Daytime: 43 schools and colleges within 1 km.
Only 3 health places within 1 km: a clinic fits the civic site.

Local architectural vocabulary



Front yard

Tiled house, Puttur



Veranda

Pillared front, Melukote



Roof form

Clay-tile roofs, Krishnarajapet



Street edge

Jagali steps, Malavalli

Raised, shaded thresholds — jagali and veranda — mediate between the street and the home, and suit Kogilu's hot April and westerly monsoon wind.

Wikimedia Commons, CC BY-SA 4.0: Prof tpms (Melukote, Krishnarajapet, Malavalli); Bharathesha Alasandemajalu (Puttur) / Old Mysore and coastal Karnataka examples, not on-site records

DESIGN / DEVELOPMENT IMPLICATION

Give each block a raised, shaded front threshold onto the internal street: a contemporary jagali that also keeps entrances above the monsoon runoff.

Networks, connections & gaps

Site 3 — Kogilu / Kogilu, Yelahanka, Bengaluru



SOURCE / Overture Maps infrastructure and roads (OpenStreetMap-derived); BBMP primary storm drains (SWD 2022). Network capacity needs provider records.

Service register

Service	Connection / distance	Capacity / status	Evidence / next check
Water	Cauvery Stage V area; mapped water tank 1.4 km W	Kogilu supply not confirmed; aquifer over-exploited	BWSSB: confirm Stage V main
Power	Substation 1.4 km W; transformer 1.1 km	Load not confirmed; 220/66 kV Yelahanka grid	BESCOM: feasibility letter
Drainage	Outfall NW → west stream → drain 284 m SW	Capacity unknown; local flood history	GBA SWD: invert levels
Sewer / waste	No sewer mapped at boundary; Jakkur STP 10 MLD	Connection not confirmed	BWSSB / BSWML: sewer, collection

DESIGN / DEVELOPMENT IMPLICATION

Water and sewer gate the delivery sequence. Until BWSSB confirms a Cauvery Stage V main and sewer, budget an on-site STP and treated-water reuse, and phase occupancy behind those approvals.



What matters for this project

Site 3 — Kogilu / Kogilu, Yelahanka, Bengaluru

01

West stream and valley buffer

Evidence p03, p07 394–5,750 m² (0.9–13.1%) may be non-buildable; merge it with the 10% park.

Next: drain class from GBA (SWD) + revenue sketch Owner: client surveyor Due: before layout freeze

02

Height clearance at 15 m

Evidence p03 Yelahanka AFS 3.2 km N, Jakkur aerodrome 2.0 km SSW, KIA 15.2 km NE: AFS NOC and AAI NOCAS for 15 m or taller.

Next: NOCAS application with top elevation ($\approx 900 \text{ m} + H$) Owner: architect Due: concept stage

03

Existing layout, water and sewer

Evidence p10, p11 A plotted layout is under construction; confirm its sanction and khata, and the Cauvery V and sewer connections.

Next: title and approvals search; BWSSB letters Owner: client legal / services Due: before acquisition

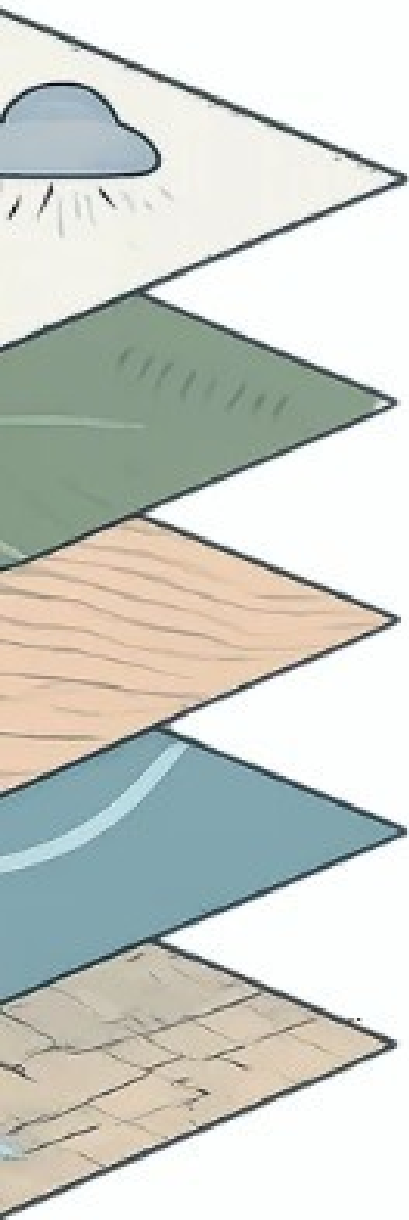
Evidence & coverage

Analysis	Dataset / date	Method / resolution	Status / limitation
Site / controls	KML Oct 2026; RCCMS; RMP 2015 ZR; draft RMP 2031 layer	UTM 43N; plan raster z15 sampling	Use confirmed; FAR from ZR text, confirm with BDA
Terrain / water	Copernicus GLO-30; Overture water; BBMP SWD 2022	30 m DSM; D8; Gumbel 1981–2025	DSM includes trees; no site survey
Climate / solar	NASA POWER MERRA-2, CERES 2015–24; NOAA ISD VOBL	0.5° / 1° grid; station 15.2 km	Regional, not microclimate
Context / services	Overture 2026-09; Esri Wayback 2017 / 2026	Walk 4.8 km/h network; 1.2 km section	Heights untagged; capacity unconfirmed

Not run / unavailable: soil texture (SoilGrids returned 503); Open-Meteo and OSM Overpass (host blocked, so NASA POWER and Overture used); Census 2011 village count; site borehole and topographic survey.

Prepared by Qnit analysis · GeoAce Studio Reviewed by: pending

Analysis run S03 / 2026-10-10 03:38 UTC Boundary version Case_Studies.kml (Oct 2026)



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design
starts
with
you.**

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